

Reverse-Engineering Mathematical Reasoning from OpenAI’s 2026 Mathematics Corpus: Implications for Mechanistic Interpretability

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Abstract

OpenAI’s public 2026 mathematics release is an unusually rich public record: mathematical manuscripts, source artifacts, Lean materials, and ten abridged reasoning summaries. It is not a release of model weights, activations, full prompts, or complete internal reasoning. We present a reproducible, preregistered artifact audit at a pinned release commit. We find 372 result families and 719 catalogue-linked manuscripts, with every linked PDF and TeX-source directory present in the Git tree. We identify 242 families with a Lean scope document and 10 released abridged summaries. Catalogue-linked manuscript count does not meaningfully differ by scope-document availability (Mann–Whitney $p = 0.638$; permutation $p = 0.665$; rank-biserial $r = 0.03$). These observations do not reverse-engineer neural mechanisms. To make that boundary operational, we separately conduct a causal residual-stream patching simulation in 4 fully inspectable toy Transformers. That surrogate result demonstrates an intervention method with seed-level uncertainty, not a property of an OpenAI model. We release code, provenance hashes, derived data, figures, and an arXiv-ready source package.

1 Introduction

Reasoning traces expose intermediate claims, proof organization, and references to verification. Yet a persuasive chain of thought is not automatically a faithful account of the internal computation that produced an answer [12, 24]. Mechanistic interpretability requires internal access and causal intervention, as in work on sparse features and circuit-level representations [7, 23].

OpenAI’s October 2026 public mathematics repository contains manuscripts and supporting artifacts produced by an unreleased internal model [21, 22]. It is valuable for studying a public record of model-assisted mathematics, but it should not be read as a transparent window into neural computation. We provide a pinned inventory, a preregistered artifact-level comparison, and descriptive measurements of the published summaries. We then separately demonstrate a genuine causal mechanistic-interpretability intervention in a toy model.

2 Scientific context: mechanistic interpretability

Mechanistic interpretability seeks explanations expressed in a model’s actual computations: features, states, edges, and algorithms that can be interrogated and causally manipulated. The circuit perspective developed from feature visualization and transformer-circuit analyses [6, 18, 19] is deliberately stronger than a useful behavioral description. It asks whether a proposed mechanism predicts the effects of intervention.

There are instructive examples in accessible models. Work on induction heads, modular-addition grokking, indirect-object identification, and a GPT-2 greater-than capability combines structure, activations, and interventions to propose circuit-level accounts [11, 17, 20, 26]. Those successes are neither a license to infer circuits from prose nor a guarantee that a localized explanation is complete. They instead specify the kind of evidence missing for an unreleased model.

Causal tools make the distinction concrete. Causal mediation, activation and path patching, model editing, automated circuit discovery, and causal scrubbing each test particular counterfactual claims [3, 4, 9, 10, 16, 25, 28]. They also expose failure modes: an ablated computation can be compensated by others, and a single intervention can overstate a component’s explanatory role [1, 8, 15].

Feature-level approaches address the fact that individual neurons can mix several concepts. Toy superposition models, dictionary learning, and sparse autoencoders supply hypotheses about more useful feature bases [2, 5, 7, 23]. In parallel, chain-of-thought and process-supervision research clarifies why visible reasoning can aid evaluation while remaining distinct from the latent computation that determined an answer [12–14, 24, 27]. Our simulation adopts the causal-intervention side of this program while making no claim to have recovered an arithmetic circuit.

3 Evidence boundary

The release contains public outputs and supporting artifacts. It does not release weights, activations, attention patterns, optimizer state, complete prompts, training data, or complete hidden reasoning. Therefore, our strongest permissible claims concern availability, organization, and surface-text statistics. We cannot identify a circuit, feature, attention head, latent representation, or causal sequence in the OpenAI model. Nor can a manuscript or a formalization scope page establish the correctness of a mathematical claim. This distinction matters because process supervision can make intermediate steps useful targets without making prose a complete causal explanation [13].

The title’s implication for mechanistic interpretability is methodological. The public record can motivate frozen benchmarks and hypotheses for inspectable models, such as whether a proof segment is causally necessary for a conclusion. It cannot answer that question without a model whose internal states can be recorded and manipulated.

4 Corpus and method

4.1 Frozen source

We acquired the Apache-2.0 licensed `openai/math` repository at commit `fd4aeeb2...0529eeeb`; the acquisition manifest records the full 40-character identifier. The manifest records SHA-256 hashes of `README.md`, `CONTENTS.md`, and `LICENSE`, plus a successful `git fsck`. Windows path limits prevent materialization of a few extremely deep upstream paths in a normal working tree. Rather than silently lose them, the pipeline reads immutable Git blobs directly from the verified object database.

The pre-analysis contract is in `RESEARCH_PROTOCOL.md` and `PREREGISTRATION.md`. The unit is a result family for catalogue measures and a manuscript for file availability. Families are not independent model runs.

4.2 Measurements

We parse each numbered family and each linked PDF in `CONTENTS.md`. For each manuscript, we measure presence of the linked PDF, a non-vendor TeX file, and a README marker. For each family, we measure whether `lean/docs/NNN.md` exists. This is Lean scope-document availability, not verified correctness or successful compilation. For the ten summary PDFs named in the upstream README, we extract pages, word-like tokens, and transparent keyword counts. Failures remain in a warnings table.

The sole preregistered comparison asks whether catalogue-linked manuscript count differs between families with and without a Lean scope document. We report medians, a two-sided Mann–Whitney test, rank-biserial correlation, and a two-sided seed-20261009 label-permutation calculation using 10,000 draws. The latter is a reference calculation, not a claim of independent random families.

5 Results

5.1 Release inventory

Artifact	Count
Result families	372
Catalogue-linked manuscripts	719
Families with Lean document	242
Linked manuscript PDFs present	719
Manuscripts with TeX source	719
Released abridged summaries	10

The parser recovered 372 families and 719 linked manuscripts, matching the upstream catalogue headline. Every linked PDF and manuscript directory with TeX sources was present in the pinned Git tree (Figure 1). There are 242 families with a Lean scope document (65.1%). OpenAI separately reports approximately 42% of top-line results as formalized [22]; this uses a different unit and is not interchangeable with our scope-document measure.

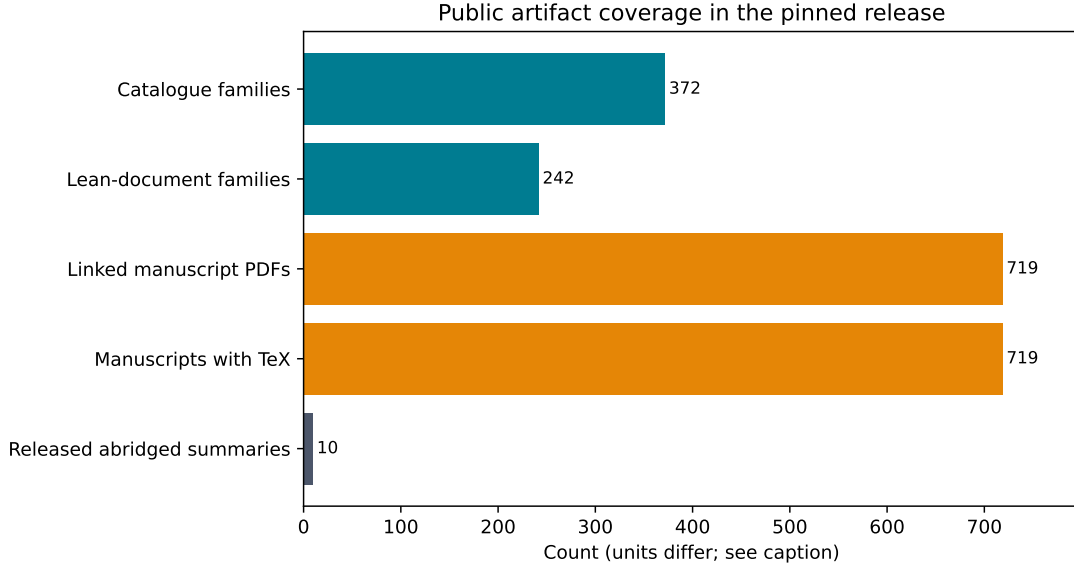


Figure 1: Public artifact coverage in the pinned release. Bars mix family, manuscript, and summary units, so they describe availability rather than a common rate or quality score.

Family size is concentrated at one linked manuscript, with a smaller tail of families containing companion arguments or consequences (Figure 2). This is a catalogue-structure observation, not a count of independent solution attempts.

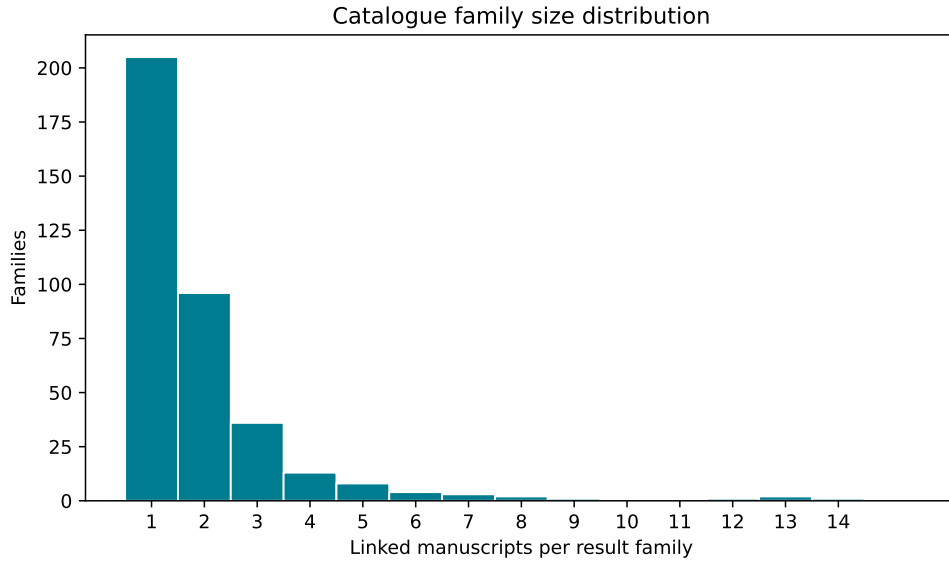


Figure 2: Distribution of catalogue-linked manuscripts per result family.

5.2 Scope documents and family size

Both groups have a median of one linked manuscript. The comparison is small and non-significant ($U = 16,150$, $p = 0.638$, $r = 0.03$); the deterministic permutation calculation gives $p = 0.665$. This does not demonstrate equality, causal independence, mathematical quality, or Lean compilation

success. It only gives no meaningful evidence that this simple catalogue-size attribute separates families by scope-document availability at this release.

5.3 Released summaries

The 10 named summaries span 5–45 PDF pages and 2,020–19,935 extracted word-like tokens (Figure 3). Their non-random selection and heterogeneity make them unsuitable for estimating properties of all model reasoning. They can support qualitative study of the published text but cannot recover omitted deliberation or establish faithfulness to unobserved activations.

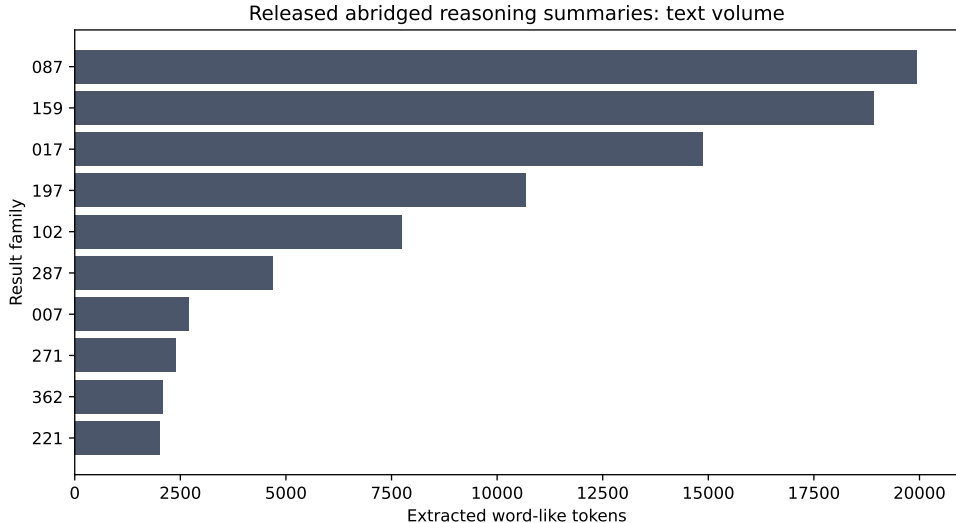


Figure 3: Extracted word-like token volume of the ten explicitly released abridged summaries. This is descriptive parser output, not a reasoning-quality or faithfulness metric.

6 Causal intervention in an inspectable surrogate

To distinguish output analysis from a mechanistic experiment, we preregistered the extension in `SIMULATION_PROTOCOL.md` before running 4 independently initialized 39,264-parameter, two-layer, three-head Transformers from scratch on all 100 ordered pairs of addition modulo 10. Their frozen vocabulary has digits, a plus token, and an equals token; every seed reached 100% training accuracy. This deliberately small, in-sample task is not a benchmark of mathematical reasoning or generalization.

For each clean prompt $(a, +, b, =)$, we corrupted the left addend with $a + 1 \pmod{10}$. At each layer and position, we replaced the corrupted post-block residual vector with the clean vector and measured recovery of the clean answer logit relative to clean and corrupted baselines. This is an activation-patching intervention [9, 28]. For prompt i , recovery is

$$R_i = \frac{\ell_y(\text{patched}, i) - \ell_y(\text{corrupt}, i)}{\ell_y(\text{clean}, i) - \ell_y(\text{corrupt}, i)}.$$

Near-zero denominators are excluded under the fixed protocol. We average R_i within model and use the four initialization seeds as the uncertainty unit, reporting two-sided t intervals with three degrees of freedom.

Layer 1 left-addend patching recovered 0.36 of the clean-logit gap (95% CI [0.20, 0.52]), compared with 0.00 (95% CI [0.00, 0.00]) for the contextual operator. The preregistered paired difference is 0.36 (95% CI [0.20, 0.52]). Layer 2 equals-position patching recovered 1.00 (95% CI [1.00, 1.00]; Figure 4). The last result is an outcome-position positive control because the unembedding reads that final residual position; it is not evidence for a distributed arithmetic circuit.

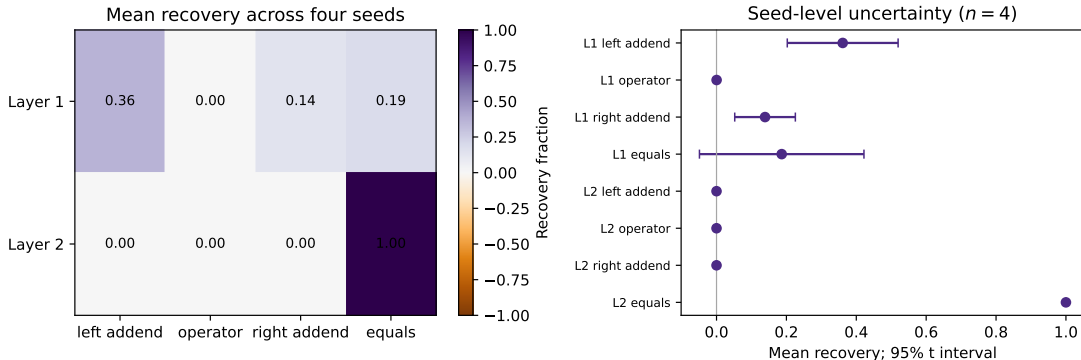


Figure 4: Four-seed causal residual-stream patching in the inspectable toy Transformer. The heat map gives mean clean-answer-logit recovery after inserting a clean residual state into a left-addend-corrupted run; points show the mean and two-sided 95% Student- t interval across four independent initializations. This figure is separate from the OpenAI artifact audit.

The surrogate result demonstrates a repeatable causal-intervention procedure. It does not establish a reusable arithmetic circuit, necessity or sufficiency of any site, generalization beyond its training table, or any property of OpenAI’s model.

7 Limitations and research agenda

This audit concerns one release commit; catalogue labels are upstream metadata, not independent adjudication. We did not compile the substantial upstream Lean project or assess mathematical validity. PDF extraction is imperfect, although failures are recorded. The toy Transformer was trained on its evaluation table, so its result is methodological rather than evidence of abstract reasoning. Most importantly, public output artifacts cannot identify mechanisms inside an unreleased model.

A credible next study should choose an open-weight mathematical model, freeze prompts and scoring, collect clean and controlled-corruption runs, and test causal effects through activation patching, path patching, ablation, or feature intervention across seeds and held-out tasks. It should explicitly separate a visible rationale from both the answer and the latent state, and test proposed circuits against multi-site or resampling interventions rather than treating a single patching map as confirmation.

8 Conclusion

The pinned release offers a substantial public artifact collection: 372 families, 719 manuscripts, Lean scope materials, and 10 summaries. A reproducible audit can characterize that public record, but cannot reverse-engineer its unreleased producer. The appropriate implication for mechanistic

interpretability is a disciplined move from interesting public traces to preregistered causal tests in inspectable models.

Reproducibility and ethics

Derived outputs are regenerated by `python -m oai_math_study.run_analysis` and `python -m oai_math_study.surrogate`. The source archive excludes third-party corpus files. This study makes no claim about proprietary model internals, intent, or hidden reasoning.

References

- [1] Anthony Bau, Yonatan Belinkov, Hassan Sajjad, Fahim Dalvi, Nadir Durrani, and James Glass. Identifying and controlling important neurons in neural machine translation. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1811.01157>.
- [2] Trenton Bricken, Adly Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- [3] Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, et al. Causal scrubbing: a method for rigorously testing interpretability hypotheses. *AI Alignment Forum*, 2022.
- [4] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2304.14997>.
- [5] Hoagy Cunningham et al. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- [6] Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- [7] Nelson Elhage, Tristan Hume, Catherine Olsson, et al. Toy models of superposition. *Transformer Circuits Thread*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.
- [8] Matthew Finlayson, Aaron Mueller, Sebastian Gehrmann, Stuart Shieber, Tal Linzen, and Yonatan Belinkov. Causal analysis of syntactic agreement mechanisms in neural language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*, 2021. URL <https://aclanthology.org/2021.acl-long.144/>.
- [9] Atticus Geiger, Duligur Ibeling, Amir Zur, et al. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL <https://jmlr.org/papers/v26/23-0058.html>.
- [10] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*, 2023. URL <https://arxiv.org/abs/2304.05969>.

- [11] Michael Hanna, Ollie Liu, and Alexandre Variengien. How does GPT-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model. *arXiv preprint arXiv:2305.00586*, 2023. URL <https://arxiv.org/abs/2305.00586>.
- [12] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023. URL <https://arxiv.org/abs/2307.13702>.
- [13] Hunter Lightman, Vineet Kosaraju, Yura Burda, et al. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023. URL <https://arxiv.org/abs/2305.20050>.
- [14] Qing Lyu, Shreya Havaldar, Adam Stein, et al. Faithful chain-of-thought reasoning. *arXiv preprint arXiv:2301.13379*, 2023. URL <https://arxiv.org/abs/2301.13379>.
- [15] Thomas McGrath, Matthew Rahtz, Janos Kramar, Vladimir Mikulik, and Shane Legg. The hydra effect: Emergent self-repair in language model computations. *arXiv preprint arXiv:2307.15771*, 2023. URL <https://arxiv.org/abs/2307.15771>.
- [16] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2202.05262>.
- [17] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2301.05217>.
- [18] Chris Olah, Arvind Satyanarayan, Ian Johnson, et al. The building blocks of interpretability. *Distill*, 2018. doi: 10.23915/distill.00010. URL <https://distill.pub/2018/building-blocks/>.
- [19] Chris Olah, Nick Cammarata, Ludwig Schubert, et al. Zoom in: An introduction to circuits. *Distill*, 2020. URL <https://distill.pub/2020/circuits/zoom-in/>.
- [20] Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022. URL <https://arxiv.org/abs/2209.11895>.
- [21] OpenAI. Sharing ai progress in mathematics, October 2026. OpenAI research announcement.
- [22] OpenAI. Mathematics manuscript collection, 2026.
- [23] Adly Templeton, Tom Conerly, Jonathan Marcus, et al. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- [24] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. *arXiv preprint arXiv:2305.04388*, 2023. URL <https://arxiv.org/abs/2305.04388>.
- [25] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, et al. Causal mediation analysis for interpreting neural NLP: The case of gender bias. *arXiv preprint arXiv:2004.12265*, 2020. URL <https://arxiv.org/abs/2004.12265>.

- [26] Kevin Wang, Alexandre Variengien, Arthur Conmy, et al. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. *arXiv preprint arXiv:2211.00593*, 2023. URL <https://arxiv.org/abs/2211.00593>.
- [27] Jason Wei, Xuezhi Wang, Dale Schuurmans, et al. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2201.11903>.
- [28] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023. URL <https://arxiv.org/abs/2309.16042>.